Ze Fan

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Wesleyan University
Department of Mathematics
265 Church St., Middletown, CT
United States

Education

Wesleyan University, Middletown, CT

May 2026 **Bachelor of Arts**
GPA: 95.95/98.30; 4.00/4.00
Major: Mathematics **Minor:** Economics

Research Experiences

5. Constructive and Stable Cartan–Dieudonné and Applications to Binary Quadratic Forms over Number Fields (joint with Han Li). Submitted to *Res. Number Theory*, Oct. 2025. (preprint)

4. The Lubin–Tate Deformation Theorem and the Morava E-theory. Expository, Sept. 2025. (preprint)

3. Stable Regularity Lemmas and their Model-Theoretic Foundations. Expository, Apr. 2025. (preprint)

2. Sparse Vectors of Small Height Avoiding Hyperplanes. Poster, July 2024. (preprint)

1. Disjoint Cycles in Ordinary Multipartite Tournaments and Round-Robin Tournaments. High School, Dec. 2020. (preprint)

Academic Awards

Dec. 2025 **Phi Beta Kappa Early Induction**

May 2025 **Wesleyan University Rae Shortt Prize**
Awarded to a junior for excellence in mathematics.

May 2024 **Wesleyan University Robertson Prize**
Awarded for excellence in mathematics to a sophomore.

May 2023 **Wesleyan University Sherman Prize-Mathematics**
Awarded for excellence in first-year mathematics.

Dec. 2020 **Award of Excellence, Global Final of S.–T. Yau High School Science Award**
In the mathematics division, nine people are awarded annually for excellent original math research projects, including one Gold Prize, one Silver Prize, three Bronze Prizes, and five Award of Excellence.
Award winning paper: Paper 1.

Talks and Seminars

2. *A Constructive and Stable Cartan–Dieudonné Theorem for Binary Quadratic Forms.* Algebra Seminar, Wesleyan University, Sept. 26, 2025.

1. *Lubin–Tate Theorem and Construction of Morava E-Theory.* Participant Talks, University of Chicago math REU, Aug. 7, 2025. (slides)

[1] Updated December 4, 2025

Skills and Languages

Programming	Python; C/C++; Standard ML; R; Stata; SageMath
Languages	Mandarin (primary); English (full professional); French(professional working).

Graduate-Level Math Courses Taken

- **MATH 509, Model Theory, 2025Sp, A+**

  This is a graduate-level course for introduction to model theory. The course covers syntax and semantics of first-order logic, definability, compactness and Löwenheim-Skolem theorems, realizing and omitting types, quantifier elimination and model completeness, countable models of complete theories.

  Instructor: Prof. Patel Rehana.

- **MATH 513-514, Analysis I, 2024Fa - 2025Sp, A and A+**

  This is the first-year graduate course in real and complex analysis. One semester will be devoted to real analysis, covering such topics as Lebesgue measure and integration on the line, abstrac measure spaces and integrals, product measures, decomposition and differentiation of measures, and elementary functional analysis. One semester will be devoted to complex analysis, covering such topics as analytic functions, power series, Mobius transformations, Cauchy's integral theorem and formula in its general form, classification of singularities, residues, argument principle, maximum modulus principle, Schwarz's lemma, and the Riemann mapping theorem.

  Instructor: Prof. Felipe Ramirez and Prof. Han Li.

  Textbook: *Real Analysis: Modern Techniques and Their Applications* by G. Folland; *Complex Analysis* by M. Stein and R. Shakarchi.

- **MATH 515, Analysis II: Lie Groups, Dynamics and Number Theory, 2025Fa, Taking**

  This graduate-level course explores homogeneous dynamics and its applications to number theory, covering the following topics: Ergodicity, mixing, and equidistribution; Recurrence in homogeneous spaces and reduction theory of quadratic forms; Spectral gaps, and the rate of mixing.

  Instructor: Prof. Han Li.

- **MATH 523-524, Topology I, 2023Fa - 2024Sp, A+ and A**

  This is the first-year graduate course in point set topology and algebraic topology. The first semester provides an introduction to topological spaces and the fundamental group; topological spaces, continuous maps, metric spaces; product and quotient spaces; compactness, connectedness, and separation axioms; and introduction to homotopy and the fundamental group. The second semester focuses on algebraic topology, concentrating on the fundamental group and homology.

  Instructor: Prof. Emily Stark and Prof. Iris Yoon.

  Textbook: *Introduction to Topological Manifolds* by J. Lee; *Algebraic Topology* by A. Hatcher.

- **MATH 525, Topology II: Low Dimensional Geometries, 2024Fa, A**

  This graduate-level course covers the structure of the homogeneous geometries in dimensions two and three. It covers the isometry groups of these spaces, with a focus on the discrete subgroups of the isometry groups together with the corresponding quotient spaces.

  Instructor: Prof. Emily Stark.

  Textbook: *The geometries of* 3*-manifolds* by P. Scott.

- **MATH 543-544, Algebra I, 2023Fa - 2024Sp, A and A+**

  This is the first-year graduate course in algebra. The first semester covers group theory including Sylow theorems, and basic ring and module theory. The second semester studies commutative algebra, Galois theory, and structure of finitely generated modules over principal-ideal domains.

Instructor: Prof. David Pollack and Prof. Alex Kruckman.

Textbook: *Abstract Algebra* by D. Dummit and R. Foote; *Introduction to Commutative Algebra* by M. Atiyah and I. Macdonald.

- **MATH 545, Algebra II: Computational Algebraic Geometry, 2024Fa, A**

  This graduate-level course is a combination of several different topics of interest, including dimension theory, algorithms for computing Hilbert series of monomial ideals, and syzygies and free resolutions.

  Instructor: Prof. Cameron Hill.

  Textbook: *Ideals, Varieties, and Algorithms* by D. Cox, J. Little, and D. O'Shea.

- **MATH 546, Algebra II: Lattices and Sphere Packing, 2025Sp, A**

  This graduate course explores the geometry and arithmetic of lattices, with emphasis on applications to sphere packing problems. Topics include Minkowski's foundational results on convex bodies and linear forms, Hermite's constant and related inequalities, and connections to reduction theory. This course also studies classical root systems and their role in dense lattice packings. Additional topics may include Voronoi cells, extreme lattices, and applications to optimization and discrete geometry.

  Instructor: Prof. Wai Kiu Chan.

- **MATH 401, Individual Tutorial: Algebraic Geometry, 2025Fa, Taking**

  This tutorial provides a graduate-level rigorous introduction to algebraic geometry through the first thirteen chapters of Vakil's *The Rising Sea*, covering category theory, sheaves, schemes, their morphisms, and geometric notions such as dimension and smoothness. Hartshorne's *Algebraic Geometry* will be used as a supplementary reference.

  Instructor: Prof. David Pollack.

  Textbook: *The Rising Sea: Foundations of Algebraic Geometry* by R. Vakil; *Algebraic Geometry* by R. Hartshorne.

Undergraduate-Level Math Courses Taken

- **MATH 223, Linear Algebra, 2022Fa, A+**

  This is a first course on linear algebra. It emphasizes a theoretical approach to the material. Topics include vector spaces, subspaces, dimension, linear transformations and matrices, determinants, eigenvalues and eigenvectors, and diagonalization of linear operators. If time permits, additional topics such as inner product spaces, Hermitian and unitary transformations, and elementary spectral theory will be discussed.

  Instructor: Prof. Wai Kiu Chan.

  Textbook: *Linear Algebra Done Wrong* by S. Treil.

- **MATH 225, Introduction to Real Analysis, 2024Sp, A**

  In this rigorous treatment of calculus, topics will include, but are not limited to, real numbers, limits, sequences and series, continuity and uniform continuity, differentiation, the Riemann integral, sequences and series of functions, pointwise and uniform convergence of functions, and interchange of limiting processes.

  Instructor: Prof. Adam Fieldsteel.

  Textbook: *Elementary Real Analysis* by B. Thomson, J. Bruckner, and A. Bruckner.

- **MATH 228, Discrete Mathematics, 2022Fa, A+**

  This course is a survey of discrete mathematical processes. Students will be introduced to the process of writing formal mathematical proofs, including mathematical induction. Topics may include set theory, logic, number theory, finite fields, permutations, elementary combinatorics, or graph theory.

  Instructor: Prof. David Constantine.

- **MATH 229, Differential Equations, 2023Sp, A**

  This course is an introduction to the theory of ordinary differential equations. Topics will include existence and uniqueness theorems as well as techniques to solve systems of equations, with applications in pure mathematics and related fields such as physics, chemistry or biology.

  Instructor: Prof. Felipe Ramirez.

- **MATH 261, Introduction to Abstract Algebra, 2023Sp, A**

  This course is an introduction to abstract principles based on the special properties of the integers, rational, real and complex numbers. The course will cover general algebraic structures as well as their quotients and homomorphisms, with emphasis on fundamental results about groups and rings.

  Instructor: Prof. Cameron Hill.

  Textbook: *A first course in abstract algebra* by J. Fraleigh.